

Optimal stimulation sites are not the most affected: personalised fMRI models of Alzheimer’s disease

Cristiano Capone^{1,*}, Enza Cece², Luca Falorsi³, Andrea Ciardiello¹, Guido Gigante¹,
Evaristo Cisbani², and Maurizio Mattia⁴

¹Dipartimento di Ambiente e Salute, Istituto Superiore di Sanità, 00161 Rome, Italy

²Intelligenza Artificiale e Tecnologie Innovative per la Salute, Istituto Superiore di Sanità, 00161 Rome, Italy

³Fondazione Santa Lucia IRCCS, 00179 Rome, Italy

⁴Dipartimento di Neuroscienze, Istituto Superiore di Sanità, 00161 Rome, Italy

*Corresponding author: cristiano0capone@gmail.com

Abstract

Targeted neuromodulation requires two decisions that are still made heuristically: which site to engage, and at what amplitude. We address both with individualised digital twins of brain dynamics. For each participant a reservoir-computing model is fit to resting-state functional MRI so that its free-running output reproduces that individual’s own lagged connectivity, and we show that the fitted models are identifiable across individuals. Because the recurrent substrate is fixed and shared, and only a linear read-out is fit per person, the twins live in common coordinates: they can be compared, averaged and perturbed. Two interventions are then contrasted on the same models. Shifting a twin’s read-out toward a control template corrects the disease signature, but the required correction is intrinsically distributed and no single-node edit reproduces it. A focal oscillatory drive at the most-deviant region fails at any amplitude, whereas a drive at the region selected by its *effect on the disease discriminant* succeeds from a single site. The two site sets are almost disjoint: the most-deviant regions are the most weakly connected, and therefore poor actuators, while effective sites are well-connected cortical nodes. Optimal targets are individual, and a closed-loop controller that titrates amplitude online from the model’s own output reaches comparable efficacy at substantially lower dose. Demonstrated here on Alzheimer’s disease, the framework is a general, model-based route to individualised target selection and dose minimisation.

Introduction

Targeted and closed-loop neuromodulation is advancing quickly, but two decisions at the centre of any such intervention are still made heuristically: which site to engage, and at what amplitude. Targets are chosen anatomically, where structural damage or network disruption is greatest, and doses are fixed in advance rather than adapted to the individual receiving them. Both would be better decided on a model of the particular brain being stimulated, one that can be interrogated counterfactually before anything is delivered. That is the promise of an individualised *digital twin*: a generative model fit to one person’s own recorded activity, whose parameters can be edited and whose dynamics can be driven, so that candidate interventions are evaluated in silico rather than by trial.

Two properties decide whether such a twin is useful. It must be *identifiable* – fit to an individual and demonstrably specific to them, not a generic model with individual noise – and it must be *perturbable*, with a well-defined way to inject a drive at a chosen site and observe the consequences for the whole system. Reservoir computing supplies both. A fixed random recurrent network provides a shared dynamical substrate; only a linear read-out is fit per individual, which makes per-subject estimation tractable from short recordings and, because the substrate is common to everyone, places all fitted models in the same coordinates, where they can be compared, averaged and interpolated. The reservoir’s separate input operator makes “drive region k ” an explicit, well-posed operation rather than an ad hoc addition.

We demonstrate the framework on Alzheimer’s disease, a setting where the targeting question is live and unresolved. AD is increasingly understood as a disorder of brain networks rather than of isolated regions, in which the progressive disruption of large-scale functional interactions parallels cognitive decline^{1;2}. That the disruption is essentially *distributed* is apparent already in metabolic imaging: the inter-regional correlation of FDG-PET uptake decreases progressively, and approximately linearly, along the transition from normal aging through mild cognitive impairment to AD, so that the loss of correlation tracks proximity to disease onset rather than damage confined to a few regions^{3;4}. Resting-state functional connectivity (FC), the inter-regional correlation structure of spontaneous BOLD activity, is among the most widely studied non-invasive readouts of this disruption, and AD is consistently associated with altered connectivity, prominently within the default-mode network (DMN)^{5;6}. FC has thus emerged as a promising biomarker of Alzheimer’s disease, although its diagnostic performance and reproducibility remain insufficient for routine clinical use⁷. Although AD is generally viewed as a distributed, network-level disorder, a more basic question remains

open at the causal level: is the AD FC alteration *reducible* to a small number of regions or connections, or is it functionally irreducible to them, distributed across the network?

The answer matters well beyond diagnosis. A focal alteration would, in principle, offer a target for the emerging arsenal of network-level interventions, deep-brain, transcranial-magnetic and transcranial alternating-current stimulation, which are most effective when a small, well-defined site can be engaged⁸. A distributed alteration, by contrast, would imply that no single target suffices. This is fundamentally a causal, counterfactual question, what would happen to a patient’s connectivity if a given site were corrected, that group comparisons of observational data cannot answer. Addressing it requires an individualised, generative model of each subject’s dynamics that can be perturbed *in silico*, in the spirit of whole-brain network models and personalised brain models^{9–13}.

Reservoir computing offers a tractable route to such a model: a fixed random recurrent network, driven by a subject’s signals, provides a rich dynamical basis from which a simple linear read-out can reconstruct that subject’s activity and connectivity^{14;15}. Fitting the read-out per subject yields a compact, individualised parameterisation of the generative process, an individualised generative *surrogate* of the subject’s FC, whose weights can be inspected, compared, and, crucially, modified. This follows a broader programme in which models inferred from neural data reproduce the data’s own spontaneous spatiotemporal dynamics when simulated freely^{16;17}, and in which network models serve to study how large-scale dynamics shape function and classification¹⁸. Here we fit such models to resting-state fMRI from AD patients and cognitively unimpaired controls, and show that the closed-loop reconstructions are faithful to and identifiable from individual connectivity, including its temporal (lagged) structure.

Building on this model, we develop the paper along a single arc, from reading out the disease to intervening on it. First, we show that two complementary read-outs of the fitted models, the geometry of the read-out weights across patients (“G-space”) and the lagged FC of the model’s own reconstruction (“FC-lag”), classify AD from controls with comparable, modest accuracy, robust to the choice of classifier and dimensionality. We emphasise that functional connectivity is not the strongest diagnostic read-out available: structural markers such as regional grey-matter atrophy separate AD from controls more accurately (Supplementary Note S1). Our purpose, however, is not diagnosis but intervention, and atrophy is not a controllable variable, one cannot stimulate lost tissue back. Stimulation acts on dynamics, so the functional read-out, even as the weaker classifier, is the appropriate substrate for a study of how activity can be causally reshaped; the FC-lag discriminant is used here as a *target* for perturbation, not as a diagnostic end in itself. Second, we

use the same models as an in-silico testbed and address the stimulation question in two stages. Targeting the most-deviant sites, the subcortical and limbic regions that diverge most from the control template, fails to revert the classification at any amplitude, establishing that the disease signature is not reducible to those focal nodes. Shifting the selection criterion to the *effect on the disease discriminant*, selecting, for each patient, the site whose resonant drive most efficiently shifts the FC-lag score toward the control distribution, achieves complete reclassification from a single, personalised site, with closed-loop titration reducing the perturbation cost further. The key finding is not merely that the signature is distributed, but that the locus of maximal pathology and the locus of maximal therapeutic effect dissociate: the former is subcortical and limbic; the latter is cortical and patient-specific.

Results

Dataset and functional-connectivity structure

The cohort comprised resting-state fMRI sessions from cognitively unimpaired controls (CC) and patients with Alzheimer’s disease (AD), parcellated into 121 cortical and subcortical regions (Fig. 1A). At the level of individual time series and their pairwise correlations the two groups were broadly similar: representative BOLD traces (Fig. 1B), group-mean FC matrices (Fig. 1D,E) and session-wise mean-FC distributions (Fig. 1C) all shared the canonical resting-state network organisation, and the population covariance was low-dimensional, a handful of principal components capturing most of the variance (Fig. 1F). Group differences were subtle and spatially diffuse rather than focal: AD sessions were on average slightly less similar to the CC-group-mean FC than CC sessions were, but the two distributions overlapped heavily (Fig. 1G), so that no single connection or region trivially separates the groups. This overlap motivated the model-based approach developed below.

A subject-specific, identifiable reservoir model of FC

For each session we fit a linear read-out from a fixed recurrent reservoir driven by the individual’s signals and ran the resulting model closed-loop to generate a synthetic reconstruction (Methods). The model captured individual connectivity well: group-mean reconstructed FC matrices (Fig. 2C,D) reproduced the corresponding empirical matrices (Fig. 2A,B), and across the four group means each model FC was most similar to its own data FC, the AD model lying closest to the AD data and the CC model to the CC

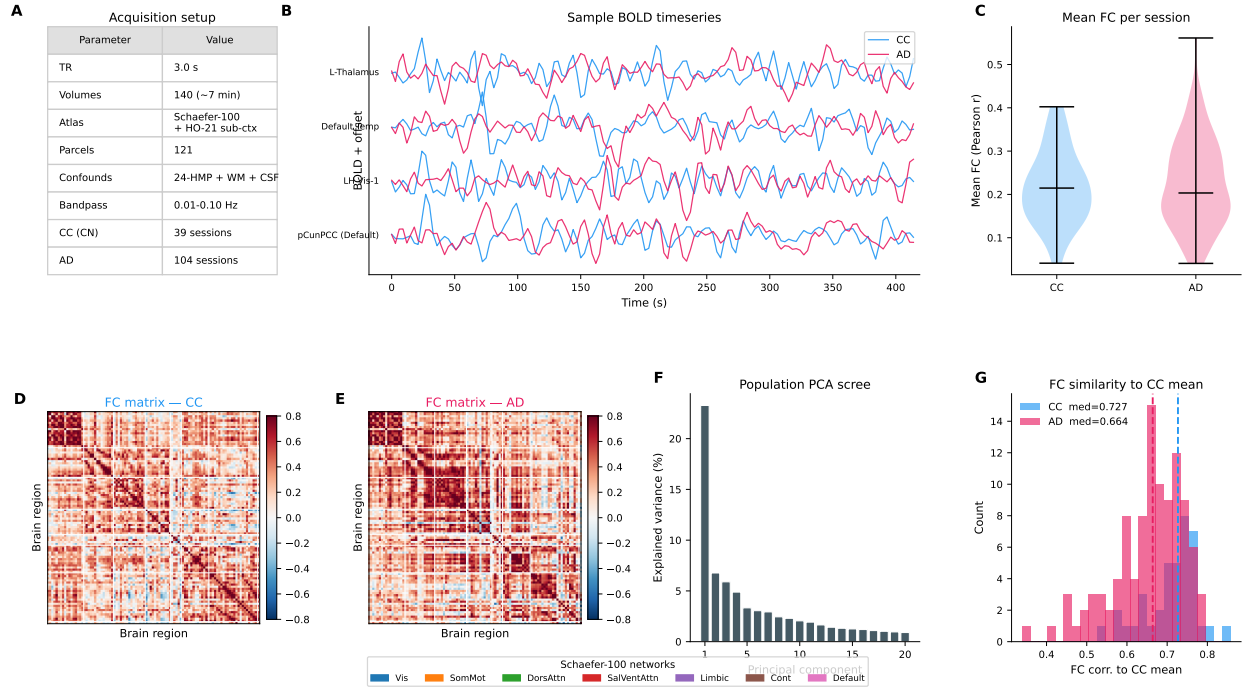


Fig. 1 | Dataset and resting-state functional-connectivity structure. (A) Acquisition and preprocessing parameters; cognitively unimpaired controls (CC) and Alzheimer’s disease (AD) session counts. (B) Representative band-pass–filtered BOLD time series for four parcels in a CC (blue) and an AD (red) session. (C) Mean functional connectivity (FC; Pearson r over parcel pairs) per session, CC vs AD. The two distributions do not differ significantly (two-sided Mann–Whitney U over all 163 CC and 104 AD sessions, $p = 0.55$; the panel shows a 39-session CC subsample), with at most a slight numerical excess in AD, consistent with the subtle, spatially diffuse group difference described below. (D,E) Group-mean FC matrices (Schaefer-100 cortical parcels, network-sorted) for CC (D) and AD (E); white lines delimit the seven Yeo networks (legend). (F) Population-PCA scree: variance explained by the leading components of the pooled parcel covariance. (G) Per-session FC similarity (Pearson r of the upper-triangular FC vector) to the CC-group mean FC, for CC and AD; dashed lines, group medians.

data (Fig. 2E). The four group-mean matrices are globally very similar (all pairwise $r > 0.93$), so this ordering is only a weak group-level effect; the decisive evidence for individual identifiability is the within- versus across-subject contrast. The reconstructions were subject-specific rather than generic: the correlation between a subject’s simulated and empirical FC was far higher within- than across-subjects (Fig. 2F,G,H), establishing that the fitted models are individually identifiable. The model also reproduced the temporal structure of connectivity, the agreement between empirical and simulated lagged FC tracking the empirical decay with delay and approaching the even/odd test–retest ceiling (Fig. 2I), with reconstruction quality peaking at an intermediate read-out regularisation and exceeding a per-session baseline (Fig. 2J). Together these results establish the model as a faithful, individualised generative account of each subject’s connectivity, the prerequisite for using it as an in-silico testbed.

Classification of AD from read-out geometry and reconstructed FC

We next asked whether the fitted models carry diagnostic information, comparing two read-outs of the same per-patient weights: the geometry of the read-out across patients (“G-space”) and the lagged FC of the model’s reconstruction (“FC-lag”) (Methods). Both separated AD from CC well above chance, but with modest and comparable performance, reaching ~ 0.66 – 0.70 AUROC (Fig. 3), in the range reported for FC-based AD classifiers on comparable cohorts^{19–22}. As a benchmark, an independent tangent-space (Riemannian) FC classifier with a linear support-vector machine, evaluated under *nested* 5×5 patient-level cross-validation with a train-only tangent reference, so that no test information enters the feature construction, reached AUROC 0.77 ± 0.16 (out-of-fold 0.76; Fig. 3, dotted band and ROC in Fig. 3E). The reservoir-based read-outs therefore lie in the same modest range as this leakage-free FC benchmark, at its lower end but well within its (wide) cross-validation spread, consistent with their being compact per-subject parameterisations rather than classifiers optimised for discrimination. As a further control we classified the *experimental* lagged FC directly, i.e. the lagged correlations of the empirical parcel BOLD (46 CC, 34 AD patients; same Gram-SVD embedding and leave-one-patient-out LDA/RF as the model-based FC-lag): this reached comparable accuracy (AUROC ≈ 0.71 , peaking at $K \approx 20$ – 25 and at lag 2), showing that reading out the *model’s* reconstructed lagged FC neither adds nor loses diagnostic information relative to the empirical lagged connectivity, but repackages the same signal in the fitted, perturbable read-out. In a two-dimensional embedding of the FC features the CC and AD point clouds overlap substantially, yet with distinct group centroids (Fig. 3F), consistent with the modest separability that every classifier attains.

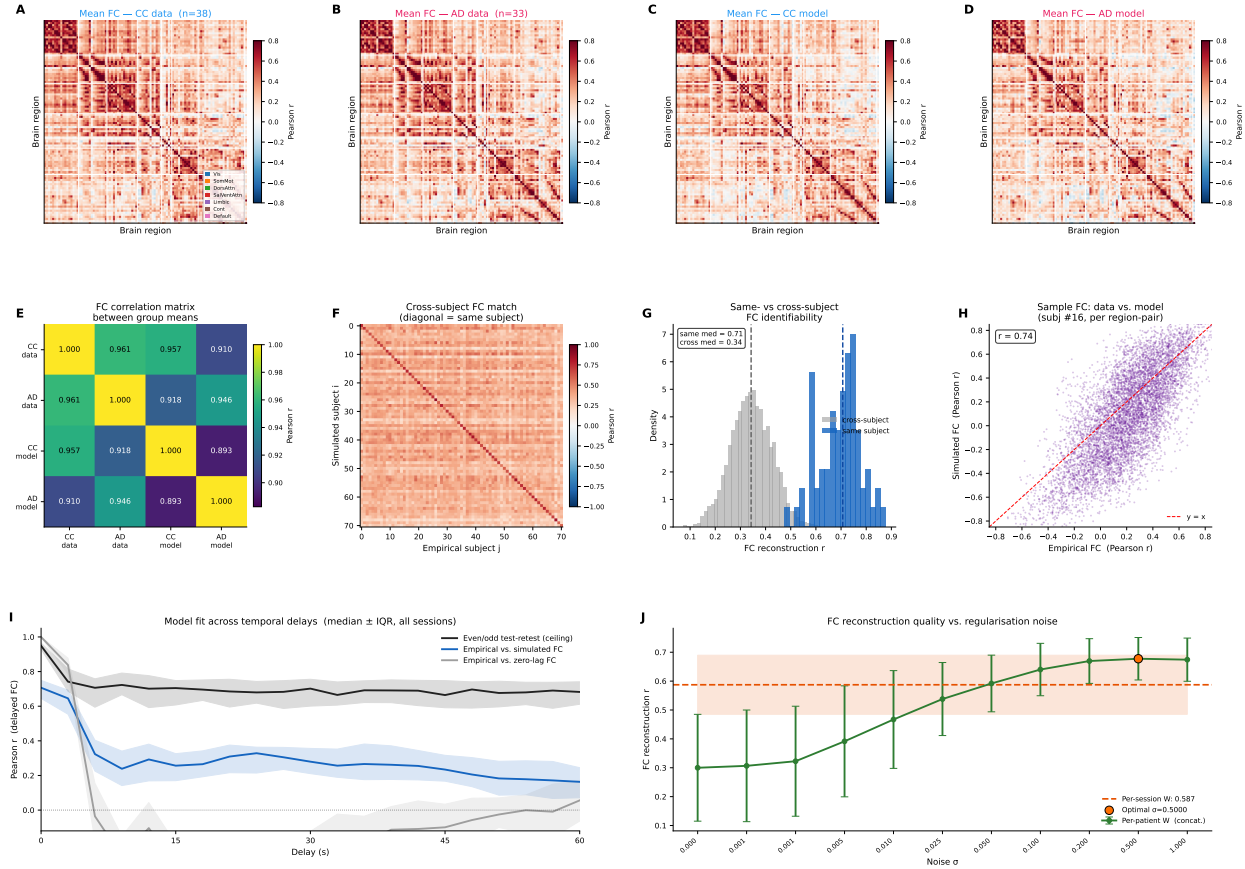


Fig. 2 | A subject-specific reservoir model reconstructs and is identifiable from individual functional connectivity. Per session, a fixed recurrent reservoir is teacher-forced on the PCA-projected BOLD; a linear read-out W is fit, and the model is run closed-loop (5-step drive, then free-running) to generate a simulated reconstruction $Y = W^T X$. (A–D) Group-mean FC for CC and AD, from data (A,B) and from the closed-loop model (C,D). (E) Correlation matrix (Pearson r) between the four group-mean FCs: each group’s model FC most resembles its own data FC. (F) Cross-subject FC matrix: correlation between simulated subject i and empirical subject j ; the diagonal (same subject) is the maximum of each row. (G) Distribution of same-subject (diagonal) vs cross-subject (off-diagonal) FC-reconstruction r ; medians dashed. (H) Empirical vs simulated FC for a representative subject, one point per region pair (r inset). (I) Model fit across temporal delays: correlation between empirical and simulated lagged FC (blue), empirical lagged vs zero-lag FC (grey), and even/odd test–retest ceiling (black); median $\pm$ IQR over sessions. (J) FC-reconstruction quality vs read-out regularisation noise σ for per-patient W (green); per-session baseline (orange); optimum marked.

Discriminability increased with the number of embedding components and plateaued near $K \approx 25$, with little benefit beyond (Fig. 3A,B), and was insensitive to the choice of classifier, a linear discriminant and a random forest performing similarly. The two feature spaces differed mainly in their sensitivity to the read-out regularisation: FC-lag was most accurate at low noise and degraded as the read-out was over-regularised, whereas G-space was comparatively noise-tolerant, remaining near its best over a broad range of σ (Fig. 3C,D). That two geometrically distinct read-outs reach a similar ceiling is itself consistent with a modest, distributed disease signal rather than a sharply localised one. Importantly, accuracy had not saturated at the present sample size: a learning-curve analysis over the number of patients used for fitting showed AUROC still rising with cohort size, and an empirical extrapolation projects ~ 0.83 AUROC at $N = 100$ and ~ 0.87 at $N = 200$ patients per group (Fig. S1), indicating that the modest accuracy reflects sample size rather than a ceiling of the representation.

In-silico stimulation: pathology site versus therapeutic target

Finally, we used the model to ask whether the AD signature can be reverted by intervention and *which* focal target, if any, makes single-site stimulation effective. We addressed this in two stages; as the analysis below establishes, the sites where the AD read-out deviates most from controls are not the sites where stimulation is most therapeutically effective. Throughout, “most-affected” sites are those with the largest per-site read-out deviation from the control template, i.e. the largest column norm $\|\Delta \mathbf{W}\|$ of $\Delta \mathbf{W} = \bar{\mathbf{W}}_{\text{CC}} - \mathbf{W}_{\text{AD}}$ (Methods). Shifting an AD read-out toward the CC template over all regions (“full-W”) moved both classifiers’ scores into the CC range at small interpolation strength and reclassified the large majority of AD patients as CC for the G-space read-out and, identically, for the FC-lag read-out (Fig. 4A,B,D,E). In sharp contrast, applying the same correction to focal site sets, the single most-affected site, the five most affected, or the most-affected site together with its ten nearest neighbours, failed to revert the classification within physiological amplitude limits, shifting only a minority of patients even at much larger interpolation strengths, again in both read-outs (Fig. 4C,F). Physiological diagnostics confirmed that the focal strategies were already operating at or beyond plausible amplitudes: whereas the full-W correction left signal amplitude near baseline, focal corrections required large local amplification to achieve even partial effects (fig. S8). An alternative focal protocol, injecting an oscillation at the most-affected sites, tuned to the reservoir’s intrinsic frequencies, showed the same limitation, single- or few-site stimulation moving the FC-lag score only weakly and only five-site stimulation at supra-physiological amplitude approaching the decision boundary (Fig. 5A). Across

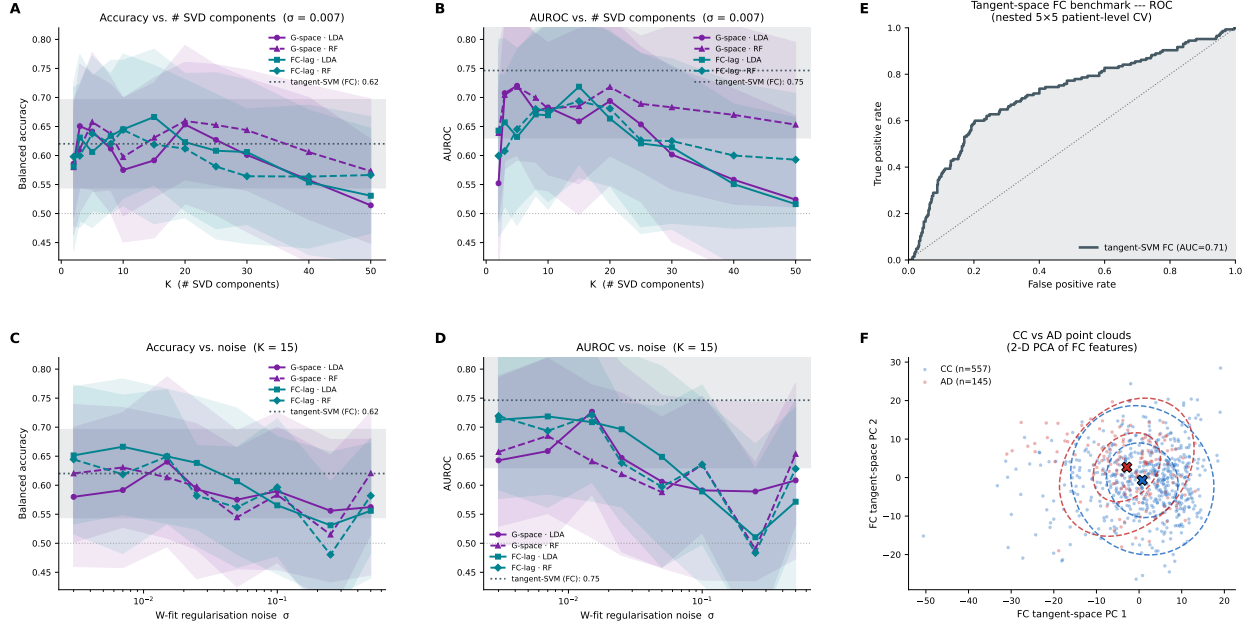


Fig. 3 | CC-vs-AD classification: read-out geometry vs reconstructed functional connectivity. Both classifiers derive from the *same* per-patient read-out W : “G-space” uses the SVD geometry of the projected W across patients; “FC-lag” uses the lagged-FC features (lags 0–2) of the reconstruction $Y = W^T X$. Each is classified with a balanced Fisher LDA and a random forest (RF), evaluated by repeated ($\times 10$) stratified 5-fold (80/20) cross-validation. Lines, mean over folds; bands, ± 1 SD; colour, feature space; line style/marker, classifier. (A,B) Balanced accuracy (A) and AUROC (B) vs the number of SVD components K (at the reference noise σ). (C,D) Balanced accuracy (C) and AUROC (D) vs read-out regularisation noise σ (at the reference K). Grey dotted line, chance (0.5); dark dotted line with band, an independent tangent-space (Riemannian) FC + linear-SVM classifier (AUROC 0.77 ± 0.16 , mean $\pm$ SD over a *nested* 5×5 patient-level CV with a train-only tangent reference and fixed, saved fold indices for reproducibility), a strong, leakage-free FC baseline. Both reservoir-based feature spaces plateau near $K \approx 25$ and reach ~ 0.70 AUROC, in the same modest range as the benchmark (at its lower end, within its wide spread), with G-space tolerating higher σ whereas FC-lag favours low σ . (E) Out-of-fold ROC curve of the nested-CV tangent-space FC benchmark (AUC ≈ 0.71). (F) Two-dimensional PCA embedding of the tangent-space FC features, coloured by group (CC blue, AD red), with 1- and 2-SD covariance ellipses and group centroids ($\times$); the distributions overlap substantially, visualising the modest but above-chance separability that every classifier reaches.

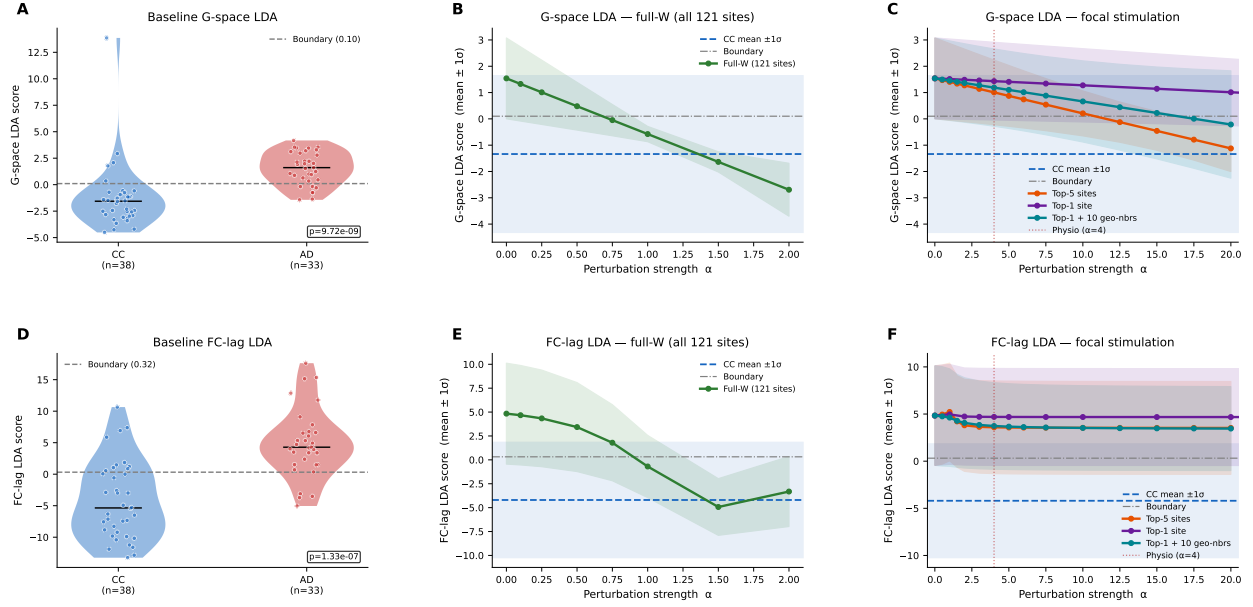


Fig. 4 | In-silico stimulation: targeting the most-affected sites fails; only full-W correction reverts the classification. AD read-outs are shifted toward the CC-mean read-out, $W_{\text{int}} = (1 - \alpha) W_{\text{AD}} + \alpha \bar{W}_{\text{CC}}$, either over all 121 sites (full-W) or over focal site sets selected by $\|\Delta W\|$ (top-5, top-1, top-1+10 geometric neighbours). (A) Baseline G-space LDA scores for CC (blue) and AD (red) with the decision boundary (dashed); two-sided Mann-Whitney $p = 9.7 \times 10^{-9}$ (also inset). (B) G-space LDA score vs α for full-W correction (mean $\pm 1\sigma$ over AD patients); CC mean $\pm 1\sigma$ shaded; full-W enters the CC range at low α . (C) Same for the three focal strategies (top-5, top-1, top-1+10 geometric neighbours); none crosses the boundary within physiological amplitude. (D–F) Analogous panels for the FC-lag LDA classifier (baseline two-sided Mann-Whitney $p = 1.3 \times 10^{-7}$). The contrast between B/E (full-W succeeds) and C/F (focal fails at any amplitude) establishes that the AD FC signature is not reducible to the most-deviant sites.

both interpolation and oscillatory protocols, no focal intervention reproduced the effect of the distributed correction, indicating that, within the fitted models, the AD FC signature behaves as functionally distributed and is not reduced to a small set of target sites.

The quantity $\|\Delta W\|$ is best understood not as a localiser of pathology but as an estimate of the change in the model's spatiotemporal connectivity kernel required to map a patient's dynamics onto the control template. As such it is intrinsically a *distributed* object: it prescribes a coordinated, multi-site, arbitrary spatiotemporal pattern of correction rather than a single lesion coordinate. Our stimulation results make this concrete. Implementing the full kernel change, interpolating all read-out columns toward the control mean, reshapes the dynamics and reclassifies the large majority of patients, whereas correcting a single column is inert (Fig. 5): the correction cannot be reduced to one node. A physically realisable focal oscillatory drive, by contrast, acts on a single input site and is effective only where the network is dynamically responsive to it, at sites selected by their effect on the discriminant, not by the magnitude of their entry in ΔW . The dissociation between the two site sets (Fig. 6) is therefore expected rather than paradoxical: a distributed

kernel correction and a focal resonant drive are different interventions, and there is no reason their leading sites should coincide.

The anatomy of the largest kernel-correction weights is nonetheless interpretable, with one caveat. The sites with the largest per-patient correction magnitude $\|\Delta\mathbf{W}\|$ were consistently subcortical and limbic, most frequently the bilateral pallidum and nucleus accumbens, the brainstem and the amygdala (Fig. 6A,B). These are, however, also among the most weakly inter-connected nodes of the parcellation, where a shared reservoir basis constrains the per-subject read-out least; the ranking should thus be read as indicating where the correction kernel carries its largest weights, not as an independently validated map of Alzheimer pathology. The dissociation is not, however, an artefact of the mixed atlas: restricted to the 100 cortical parcels alone the same relation holds ($\rho = -0.88$, $p = 5 \times 10^{-33}$), and the cortical most-affected sites and cortical targets share no parcel.

This failure of the pathology-guided approach motivated a change of selection criterion. Rather than targeting the site of maximal read-out deviation, we asked: which site, driven at the reservoir's resonant frequency, moves each patient's FC-lag score most efficiently toward the control distribution? This *discriminant-aligned* target is a fundamentally different object from a lesion marker, it is selected by its effect on the classifier, not by its own magnitude of change, and identifying it requires the per-patient generative model. The analysis below shows that this change of criterion is decisive: the same type of focal, single-site, resonant intervention that fails at the deviation-ranked target succeeds completely at the discriminant-aligned one.

Beyond identifying *where* the disease signal concentrates, the model also prescribes *how* to drive a target. Efficacy is sharply frequency-tuned: a sinusoidal drive at a focal site moves the FC-lag score toward the control distribution most efficiently when its frequency matches the reservoir's dominant eigenmode f_{eig} , the resonant drive far outperforming an off-resonance one per unit amplitude (fig. S6). This frequency, $f_{\text{eig}} \approx 0.077$ cycles/step, corresponds to ≈ 0.026 Hz once expressed in physical time through the TR (= 3 s), a unit conversion, not a dependence of the resonance on TR, placing it in the well-established infraslow band in which resting-state connectivity itself fluctuates^{23–25}, so the same fitted model specifies both a location (the subcortical/limbic targets of Fig. 6A,B) and an optimal driving frequency, a spatial *and* spectral prescription that descriptive analyses cannot provide. As amplitude grows the stimulated AD score shifts steadily toward the control range and a growing fraction of patients cross the decision boundary. A full per-patient amplitude×frequency scan confirms that reclassification concentrates in a narrow band at the

resonance f_{eig} , where patients cross the boundary at the lowest amplitude and hence the smallest departure from the unstimulated simulation, though reclassification and that departure trade off, so full reversal still demands a large perturbation (fig. S6). At physiologically plausible amplitudes a focal drive does not, by itself, revert the classification (Fig. 4); its significance is methodological, demonstrating that an identifiable generative model yields concrete, testable stimulation parameters. A direct comparison of focal protocols on a common classifier shows that the *site-selection criterion* matters far more than the number of sites: a 2-site drive matched to the two leading reservoir eigenmodes and a single-site resonant drive at the most-deviant site, selected for modal excitation and disease norm respectively, both reclassify at most $\sim 36\%$ of patients. Choosing the target instead by its *effect on the discriminant*, the site whose resonant drive moves the FC-lag score most toward CC, a criterion we call *LDA-resonant*, is markedly more effective: a single population-average such site reclassifies $\sim 48\%$ of patients, and personalising it per patient reverts nearly all of them ($\sim 94\%$) (Fig. 5A). These targets are predominantly cortical, led by the left Default-network prefrontal cortex (fig. S3A,B), and highly heterogeneous, 25 distinct sites for the 33 patients (fig. S4), so no single global target is best for all. Tellingly, the personalised LDA-resonant and the eigenmode-coupling drives perturb the simulation *near-identically* (correlation to the unstimulated model $\approx 0.32\text{--}0.34$; fig. S5), yet only the discriminant-aligned target converts that perturbation into near-complete reclassification ($\sim 94\%$ vs $\sim 36\%$): selecting the site by its effect on the discriminant extracts more therapeutic effect per unit of dynamical perturbation, and does so by perturbing a single input site rather than rewriting all 121 read-out weights. These reversions still require supra-physiological amplitudes (fig. S8); their significance is methodological, showing that an identifiable per-subject model turns “where and how to stimulate” into a directly computable prescription.

To isolate what makes a *single-site* intervention work, we contrasted the two ways of acting on one node (Fig. 5): the *theoretical* read-out correction, which interpolates that node’s read-out column toward the control mean, and the *physical* resonant drive ($A \sin(2\pi f_{\text{eig}} t)$). The single-column correction is essentially inert: even at $\alpha = 50$ it never reverts the classification (flat at the $\sim 18\%$ baseline) while progressively distorting the simulation, so a single read-out weight has negligible leverage on the distributed FC-lag signature. The resonant drive instead reclassifies by engaging the network-wide resonant mode, but *where* it is applied is decisive: at the ΔW most-affected (subcortical) site it reaches only $\sim 18\text{--}21\%$ even at high amplitude, whereas at each patient’s discriminant-aligned site it reverts nearly *all* of them ($\sim 94\%$) (Fig. 5A). Either way its distance from the original FC jumps to ~ 0.7 at the smallest effective amplitude and saturates

(Fig. 5B,C). At one site, then, a read-out edit is inert and only a resonant drive at the right site cures, at a fixed, large perturbation cost that the closed-loop control below is designed to minimise.

Having established that the therapeutic target is defined by its effect on the discriminant rather than by pathological deviation, we can compare the two site sets directly (Fig. 6C–E). Across the 33 AD patients, the 29 distinct sites ever appearing in a patient’s pathology top-5 and the 25 distinct per-patient therapy sites (personalised LDA-resonant targets) share only 1 site. The great majority of each set is exclusive to its own criterion, and the therapy sites, unlike the pathology sites, have no shared anatomical hub.

Effectiveness of state-dependent perturbation at single-subject level

Because the fitted model is a closed-loop dynamical system, the same FC-lag score that diagnoses a patient can serve as the feedback signal for a *real-time, causally valid* adaptive intervention: rather than an offline oracle with access to the full future trajectory, we titrate the resonant drive’s amplitude online, every few steps, from a sliding window of the model’s own past output (Methods). The explicit window is a deliberately conservative choice: because the reservoir satisfies the echo-state property, the relevant past is already embedded in its instantaneous state, so an equivalent controller could read the biomarker directly from $\mathbf{x}(t)$ without buffering past output²⁶. At the fixed, personalised LDA-resonant site, this online controller reclassifies 26/33 (~79%) of AD patients, approaching the ~88% reached by a fixed open-loop drive, while cutting the mean stimulation amplitude by ~31% ($A = 6 \rightarrow 4.1$) and the departure from each subject’s own dynamics by ~5% among patients reclassified under both protocols (distance $0.69 \rightarrow 0.66$; Fig. 5D–F). Online amplitude titration therefore converts the costly fixed drive into a substantially cheaper, individualised protocol that reaches nearly the same in-silico reversal closer to the patient’s own resting dynamics, using only information available causally, in real time.

A natural concern is that this reclassification could be inflated by fitting the discriminant on the same patients it scores. We therefore re-evaluated the entire pipeline under leave-one-AD-patient-out cross-validation: the FC-lag embedding, the LDA discriminant, its decision threshold, *and* the site-selection criterion were all rebuilt from the training patients only and then applied to the held-out patient (Methods; fig. S7). The therapeutic effect survives. Honest cross-validation does raise the baseline (unstimulated) misclassification of AD as CC from 21% to 39%, so that only 20 of the 33 AD patients lie on the AD side of the boundary without stimulation, consistent with the modest classifier accuracy; we therefore quantify the effect as the net fraction of these AD-side patients that stimulation moves across to the CC side. A single

Single-site stimulation and closed-loop control: a single read-out column is inert, a resonant drive cures only at the discriminant-aligned site, and feedback minimises the perturbation

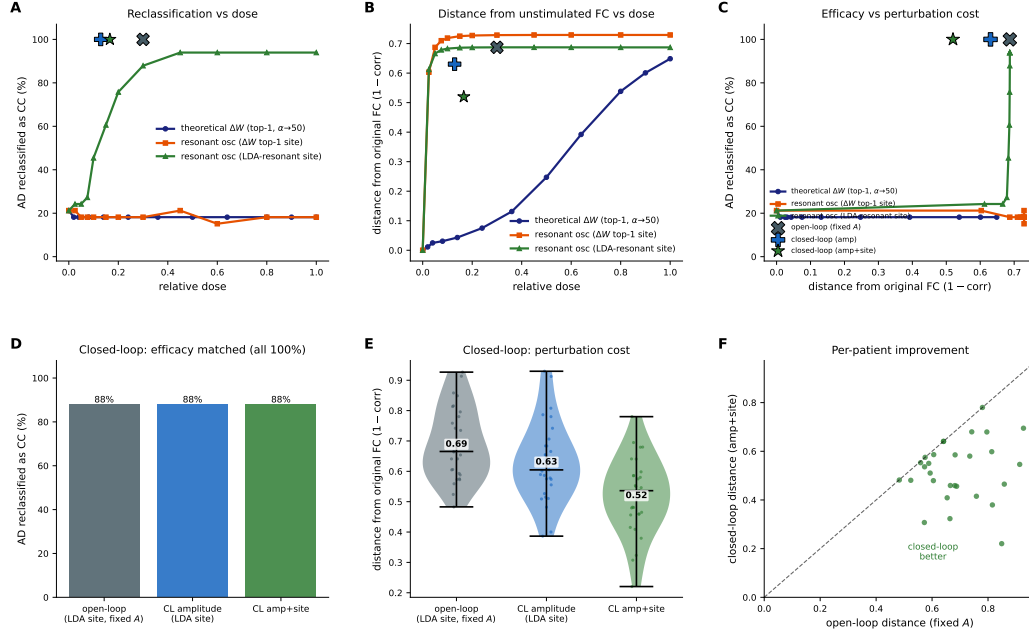


Fig. 5 | Single-site stimulation and closed-loop control ($N = 33$ AD; distance = $1 - \text{corr}$ between the stimulated and unstimulated closed-loop free-run FC). *Top row, single-site dose-response* comparing a *theoretical* read-out correction (ΔW interpolation of the top-1 most-affected column toward the CC mean, $\alpha \in [0, 50]$), a *resonant* drive $A \sin(2\pi f_{\text{eig}} t)$ at that same ΔW site, and a resonant drive at each patient's *LDA-resonant* (discriminant-aligned) site ($A \in [0, 20]$). **(A)** Reclassification vs normalised dose: the single-column ΔW correction is inert (flat at the 18% baseline even at $\alpha = 50$); the resonant drive at the ΔW site stays near ~18–21%; the resonant drive at the LDA-resonant site reverts nearly *all* patients (~94%). **(B)** Distance from the original FC: the ΔW correction distorts only gradually, whereas both resonant drives jump to ~0.7 at the smallest effective amplitude and saturate (the oscillation dominates the network-wide mode). **(C)** Efficacy vs perturbation cost: only the LDA-resonant drive converts perturbation into near-complete reclassification. The closed-loop *operating points* (patient means; grey $\times$ open-loop, orange $+$ online closed-loop) are overlaid on **A–C**: the online closed-loop point sits at a substantially smaller dose than the open-loop point, for a comparable, though lower (~79% vs ~88%), reclassification rate and a modestly smaller distance. *Bottom row, real-time closed-loop control at the LDA-resonant target* (the same curing site as the green curve of **A**), comparing a fixed *open-loop* drive ($A = 6$) to an *online closed-loop* controller that titrates amplitude every few steps from a sliding-window estimate of the FC-lag biomarker computed only from the model's own past output, no oracle access to the future trajectory (Methods). **(D)** Per-patient stimulation amplitude (dose): online control lowers the mean dose from 6.0 to 4.1 (~31% lower) while approaching open-loop efficacy (26/33, ~79%, vs 29/33, ~88%). **(E)** Distance from baseline among patients reclassified under both protocols ($n = 26$): online control lowers the mean distance by ~5% (0.69 $\rightarrow$ 0.66). **(F)** Per-patient: most subjects lie below the diagonal, i.e. real-time control reaches nearly the same reversal at a smaller perturbation cost, the in-silico analogue of adaptive (closed-loop) neuromodulation, using only information available causally, online.

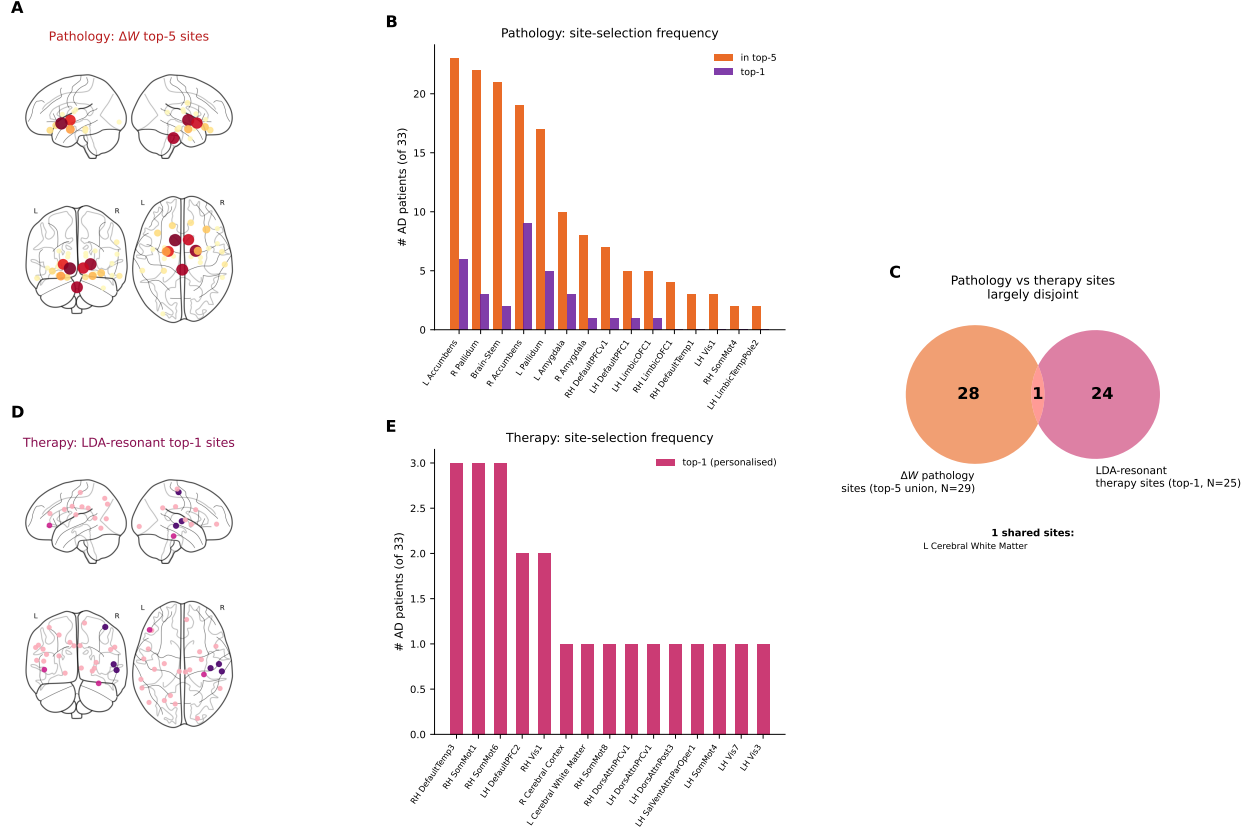


Fig. 6 | Pathology sites and therapeutic-stimulation sites are largely disjoint. *Pathology* sites are ranked per AD patient by the per-site read-out correction magnitude $\|\Delta \mathbf{W}\|$ (column norm of $\Delta \mathbf{W} = \bar{\mathbf{W}}_{CC} - \mathbf{W}_{AD}$, top-5 per patient). *Therapy* sites are each patient's personalised LDA-resonant target, the single site (top-1) whose resonant drive moves that patient's FC-lag discriminant most toward CC (Methods). **(A)** Glass-brain markers at the pathology sites (MNI centroids), coloured and sized by top-five selection frequency: deep subcortical/limbic location. **(B)** Pathology selection frequency: number of AD patients with each site among the top-five (orange) or as the single most-affected site (top-1, purple). The leading sites are subcortical and limbic, bilateral pallidum and nucleus accumbens, the brainstem and the amygdala. **(C)** Overlap between the two site sets: of the 29 distinct pathology sites (top-5 union across patients) and 25 distinct therapy sites (top-1 per patient), only 1 site ($\sim 3\%$ and $\sim 4\%$ of each set, respectively) is shared; each set is almost entirely exclusive to that criterion. **(D)** Glass-brain markers at the therapy sites, coloured and sized by top-1 selection frequency: predominantly cortical and, unlike the pathology sites, without a shared anatomical hub, each site is selected by at most three of the 33 patients. **(E)** Therapy selection frequency: the leading therapy sites are the left Default-network prefrontal cortex and visual association cortex, with no site selected by more than three patients.

population-average resonant target, selected on the training patients alone, achieves a leakage-free net cure of ~25%, whereas each patient's *personalised* LDA-resonant target moves ~90% of them across (100% at $A = 10$; Methods). The single-site reclassification is therefore a genuine, model-prescribed effect rather than an artefact of classifier leakage, and the gap between the two targets is itself the signature of target heterogeneity: no one site serves the group, which is precisely why the population-average target recovers only a minority. The personalised figure is nonetheless an upper bound, since each patient's site is chosen by argmax under the same discriminant that then scores it; the efficacy of a deployable protocol lies between the two arms and is not settled here.

Discussion

We fit identifiable, subject-specific reservoir-computing models of resting-state functional connectivity and used them both to read out and to intervene on the Alzheimer's signature. Two complementary read-outs of the fitted models, the geometry of the read-out weights across patients and the lagged FC of the model's own reconstruction, classified AD from controls with comparable, modest accuracy. The central finding of the stimulation analysis is a dissociation between the anatomy of pathology and the anatomy of effective intervention. Targeting the sites of greatest read-out deviation from controls, predominantly subcortical and limbic, fails to revert the classification even at supra-physiological amplitudes, confirming that the FC signature is not concentrated in those regions. Selecting the target instead by its effect on the disease discriminant under a resonant drive (the LDA-resonant criterion) achieves complete reclassification from a single, individualised site, identifying for each patient the node where stimulation is most therapeutically effective. The two site sets are anatomically distinct: pathology concentrates in subcortical and limbic hubs, whereas therapeutic leverage resides in cortical nodes, predominantly prefrontal default-network and dorsal-attention cortex, that propagate network-wide corrections through their connectivity. Whether this dissociation reflects the biology of AD or the representational geometry of the model is a question our framework raises but cannot, on its own, settle.

This conclusion is consistent with the view of AD as a network, or “disconnection”, disorder rather than a regionally circumscribed one^{1,2}, and extends it with a causal, counterfactual argument that observational group comparisons cannot make: within an individualised generative model, targeting where the pathology is greatest is not equivalent to targeting where stimulation is most effective. The practical implication for

neuromodulation is twofold. First, a single-target intervention *can* revert a distributed connectivity deficit, but only when the target is selected for its effect on the disease discriminant rather than for its pathological magnitude; this distinction is not accessible without a per-patient generative model. Second, the effective site is highly patient-specific (25 distinct parcels across the 33 AD patients), so no single anatomical landmark serves all individuals: personalised, model-informed targeting is required. Both the per-patient discriminant-aligned sites and the population-average LDA-resonant target (left prefrontal default-network cortex) are consistent with loci explored in non-invasive AD neuromodulation^{8;27}, lending translational plausibility to the model-derived prescription. For patients whose optimal site is inaccessible or whose response at a single site is insufficient, the present framework also provides a way to design distributed, multi-site or network-shaped stimulation strategies in silico before any intervention, complementing model-based approaches that predict stimulation spread in whole-brain models^{28–31}.

The anatomy of the read-out deviations is suggestive, though it should be read with care. The regions whose read-out departed most from the control template were predominantly subcortical and limbic, bilateral pallidum and nucleus accumbens, brainstem and amygdala, structures of the limbic and basal-ganglia circuitry; several of these overlap with regions harbouring early neurofibrillary pathology in Braak staging^{32;33}, although they are also the most weakly inter-connected nodes of the parcellation, where the per-subject read-out is least constrained, so this correspondence should be treated as provisional rather than as a validated disease localiser. Regardless of that interpretation, these maximally-deviant regions were not viable focal intervention targets: the site of the largest single-node correction and the site of greatest therapeutic leverage dissociate cleanly. A plausible mechanistic account is that pathology originating in limbic hubs propagates along structural and functional connections, distributing its imprint on connectivity across the network^{34;35}; the discriminative signal then resides in the network-wide pattern, so that correcting the seed regions alone is insufficient. The cortical nodes that do provide therapeutic leverage, prefrontal default-network, dorsal-attention and visual cortex, are precisely those best positioned to redistribute a corrective perturbation network-wide through their dense long-range connectivity, and they interface with the subcortical structures of greatest pathological deviation via established DMN–limbic pathways. This dissociation between lesion locus and effective intervention site is consistent with the network-medicine principle illustrated most clearly for deep-brain stimulation⁸: the optimal target is not the damaged node but the network node most effectively coupled to the symptom circuit. The per-patient heterogeneity of the optimal sites (25 distinct parcels for 33 patients, Fig. S4) further argues that no universal anatomical landmark suffices, a result consistent

with the clinical heterogeneity of AD and with the emerging literature on connectivity-guided personalised neuromodulation.

A distinctive feature of the framework is that it prescribes not only where but how to stimulate. The driving frequency that most efficiently moved a target toward the control distribution coincided with the model’s intrinsic resonance (fig. S6), connecting our results to the broader principle of frequency-specific neuromodulation. Stimulation efficacy is well known to depend on frequency, and entraining endogenous rhythms is an active therapeutic strategy in AD, most prominently 40 Hz “gamma” sensory and electrical stimulation, which reduces pathology and improves function in mouse models and early human studies^{36;37}. Our prescription operates at a far slower, infraslow timescale, the band in which resting-state connectivity fluctuates, rather than in the gamma range, reflecting the BOLD-derived nature of the model; we therefore make no claim about a specific deliverable protocol. The conceptual contribution is that an identifiable, subject-specific model turns “at what frequency should one stimulate?” into a computable quantity tied to the network’s own dynamics, complementing connectome-model approaches that predict the spatial spread of stimulation^{28;30}. A practical advantage is that, because the reservoir dynamics and the read-out are *linear*, the response to a periodic drive, and hence the optimal stimulation site and frequency, follows directly from the system’s transfer function and eigenspectrum, rather than from an exhaustive search: the resonance we observe is precisely the peak of this linear frequency response. The model thus affords a closed-form, analytically tractable prescription of optimal stimulation parameters. Selecting the target by its *effect on the discriminant under a resonant drive* (the LDA-resonant criterion) jointly resolves drivability and direction, and is why it outperforms modal-coupling or disease-deviation selection. Reassuringly, when the target is selected for its effect on the disease discriminant, the leading “drive-toward-health” site falls in the Default-network prefrontal cortex (fig. S3), a hub of the network most consistently disrupted in AD^{5;6} and a prime locus for non-invasive neuromodulation, e.g. precuneus/Default-network stimulation trials²⁷, lending biological plausibility to the model-derived prescription. The heterogeneity of the per-patient optimal targets further argues that effective targeting must be individualised rather than one-size-fits-all, consistent with connectivity-guided personalised stimulation⁸. Because the fitted model is a dynamical system, the same biomarker that scores a patient can also *close the loop* in real time. Titrating the drive’s amplitude online against a sliding-window FC-lag estimate, using only the model’s own past output rather than an offline oracle, reclassifies ~79% of patients (approaching the ~88% open-loop rate) while cutting both the stimulation dose (~31%) and the departure from their own dynamics

(~5% among patients reclassified under both protocols) relative to a fixed open-loop drive (Fig. 5). This is the in-silico counterpart of adaptive (closed-loop) neuromodulation, which seeks a minimal, causally available perturbation that crosses the diagnostic boundary rather than a supra-physiological open-loop drive. We emphasise, however, that these are *model-level* prescriptions: the reservoir is fit to infraslow BOLD connectivity, not electrophysiology, so an “intervention” here is an abstract input to the regional FC dynamics, and its optimal spatial pattern and (infraslow) frequency should be read as targets at the level of regional drive, not as cellular or oscillatory mechanisms. With that caveat, the focal, slowly-modulated character maps in spirit onto closed-loop, biomarker-guided neuromodulation; translating it to any specific technology, and validating it against electrophysiological or longitudinal outcomes, is left to future work.

More broadly, the study illustrates a general methodology: an identifiable individualised generative surrogate of a subject’s dynamics, fit once, can serve simultaneously as a biomarker substrate and as a counterfactual testbed. Because the read-out is a compact per-subject parameterisation, the same fitted object that classifies the disease can be perturbed to ask what would reverse it, a loop from decoding to intervention that is difficult to close with descriptive analyses alone, and that transfers directly to other conditions and to stimulation planning.

Several limitations qualify these conclusions. The cohort is cross-sectional and of modest size, and the classifiers, while honestly cross-validated, reach only moderate accuracy and embed each subject relative to the group (a transductive embedding); per-subject read-outs were estimated from limited data, adding fitting noise that we characterised but did not eliminate. The therapy-target reclassification is not, however, an artefact of this transductive fit: it survives a fully leave-one-AD-patient-out reconstruction of the embedding, discriminant, threshold and target selection, once the effect is read as the net fraction of AD-side patients moved to control, above an honest (higher) baseline (fig. S7). The reservoir is a phenomenological rather than biophysical model, with fixed random recurrent weights, and our “stimulation” is an idealised modification of the read-out or its input rather than a literal physical protocol; the mapping between in-silico amplitude and a deliverable dose is therefore approximate, and the anatomical neighbourhoods rely on coarse parcel centroids. These factors bear on quantitative details but not on the central, qualitative contrast between the distributed and focal interventions, which is large and consistent across classifiers and protocols.

Future work should test the distributed signature longitudinally and in larger, multi-site cohorts, replace the phenomenological reservoir with more biophysically grounded generative models, and connect the in-silico targets to realisable stimulation montages so that distributed and multifocal strategies can be evaluated

explicitly. Formally testing whether the distributed FC signature follows the predicted spread of pathology along network connections, rather than merely co-localising with the most-affected hubs, would turn the reconciliation proposed above into a falsifiable hypothesis.

Methods

Data and preprocessing. Resting-state functional MRI sessions were downloaded from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu), comprising cognitively unimpaired controls (CC) and patients with Alzheimer’s disease (AD). Each session comprised 140 volumes (repetition time $TR = 3.0$ s, ~ 7 min). Images were minimally preprocessed and parcellated into $N_p = 121$ regions of interest, the 100 cortical parcels of the Schaefer 7-network atlas³⁸ plus 21 Harvard–Oxford subcortical labels. All 121 parcels (cortical and subcortical) enter the reservoir fit, the classifiers and the stimulation analyses throughout; the network-sorted group-mean FC matrices of Fig. 1D,E display the 100 cortical parcels only, for which the Yeo 7-network assignment is defined. Region-wise BOLD time series were denoised by regression of 24 head-motion parameters and mean white-matter and cerebrospinal-fluid signals, and band-pass filtered to 0.01–0.10 Hz. To balance the two groups during reservoir fitting, classification and stimulation we sampled 40 CC sessions from the 48 verified CC subjects; grouping by subject (one session each) and applying the volume filter gives 38 CC and 33 AD subjects. The full pool (46 CC and 34 AD subjects; 163 and 104 sessions) was used for the cohort-size scaling and for the tangent-space SVM benchmark. Diagnostic labels come from ADNI’s own diagnosis query and are independently corroborated by cognition: matching each scan to the nearest MMSE visit (median gap 15 days) gives 28.7 ± 1.4 for CC and 22.7 ± 2.8 for AD (two-sided Mann–Whitney $p = 1.6 \times 10^{-13}$), and the FC-lag discriminant is unrelated to age (Spearman $\rho = +0.10$), education ($\rho = -0.09$) or sex ($\rho = -0.16$; all $p > 0.15$), so the separation is not a demographic artefact. The latter used the globally signal-regressed sessions (163 CC + 104 AD sessions passing a >100 -volume filter) under *nested* 5×5 subject-grouped cross-validation: Ledoit–Wolf covariances projected to the tangent space at a *train-only* reference mean, then standardised, PCA-reduced, and classified by a linear SVM or logistic regression, with the embedding dimension, penalty and classifier chosen in the inner loop and evaluated on held-out subjects in the outer loop. Sessions were grouped by subject, and per-subject quantities used the first available session unless otherwise stated; Table S1 summarises the cohort used in every analysis.

Population principal-component projection.

All region time series were pooled and mean-centred, and the leading $N_{\text{PC}} = 50$ eigenvectors $\mathbf{V}_{50}$ of the $N_{\text{p}} \times N_{\text{p}}$ covariance were retained. For each session with signal $\mathbf{s} \in \mathbb{R}^{N_{\text{p}} \times T}$ we defined the PCA-reconstructed *target* $\hat{\mathbf{s}} = \mathbf{V}_{50} \mathbf{V}_{50}^T \mathbf{s}$, which removes idiosyncratic high-order variance while preserving the dominant spatiotemporal structure. Empirical functional connectivity (FC) was the Pearson correlation matrix of $\hat{\mathbf{s}}$. The $N_{\text{PC}} = 50$ projection defines only the reservoir’s training *target*; the read-out and all FC quantities remain in the full $N_{\text{p}} = 121$ region space, so the regional dimensionality is never reduced below 121.

Reservoir model.

We used a fixed random recurrent network of $N = 2000$ rate units with state $\mathbf{x}(t)$, internal weights $\mathbf{J}$ (Gaussian, rescaled to spectral radius 0.95), and fixed random input weights $\mathbf{J}_{\text{in}}$ ($\sigma_{\text{in}} = 0.01$; the subscript “in” distinguishes these untrained input weights from the trained read-out $\mathbf{W}$). The network is a *linear*, deterministic reservoir: the update below is linear in $\mathbf{x}$ and no endogenous noise is injected into the dynamics (regularisation noise enters only the read-out fit, Eq. 2). Units evolved in discrete time ($dt = 0.005$) as

$$\mathbf{x}(t + dt) = e^{-dt/\tau} \mathbf{x}(t) + (1 - e^{-dt/\tau}) (\mathbf{J} \mathbf{x}(t) + \mathbf{J}_{\text{in}} \mathbf{u}(t)), \quad (1)$$

with a short membrane time constant τ (effectively instantaneous update), and a linear read-out $\mathbf{y}(t) = \mathbf{W}^T \mathbf{x}(t)$ in the $N_{\text{p}} = 121$ -dimensional region space. The same reservoir (identical $\mathbf{J}, \mathbf{J}_{\text{in}}$) was reused for every session.

Why a reservoir rather than an autoregressive model.

A per-subject generative model of $N_{\text{p}} = 121$ regions must be estimated from a single session of ~ 140 volumes, and that ratio, not a preference for any particular architecture, dictates the choice. A subject-specific vector autoregression would require N_{p}^2 coefficients per lag: at one lag that is 121 predictors per region estimated from 139 samples, already at the edge of identifiability, and the temporal structure of interest spans several lags, where the problem is plainly underdetermined. The reservoir moves that burden off the data. Its recurrent and input operators are fixed and *identical for every session*, so the only quantity estimated per subject is a ridge-regularised linear read-out on a common state basis, whose effective degrees of freedom are set by the regularisation rather than by the raw parameter count. The distribution of timescales that a vector autoregression would have to buy with many lags is instead supplied by the reservoir’s own dynamics.

Two further properties matter for what follows, and neither is available from a fitted autoregression. Because $\mathbf{J}$ and $\mathbf{J}_{\text{in}}$ are shared, read-outs from different subjects are expressed in the same coordinates,

so differences between them ($\Delta \mathbf{W}$), group averages ($\bar{\mathbf{W}}_{CC}$) and interpolations between individuals are meaningful operations rather than comparisons of separately parameterised systems. And the reservoir carries an explicit input pathway $\mathbf{J}_{in}$ that is distinct from its recurrent dynamics, so injecting a drive at one region is well defined and separable from the model’s intrinsic behaviour. Both are prerequisites for the perturbation analyses reported above.

Read-out fitting. For each session the reservoir was teacher-forced by feeding the target as input, $\mathbf{u}(t) = f \hat{\mathbf{s}}(t)$ with feedback gain $f = 0.1$, and the hidden states $\mathbf{X} \in \mathbb{R}^{T_{eff} \times N}$ were collected after discarding the first 10 steps. The read-out $\mathbf{W} \in \mathbb{R}^{N \times N_p}$ was obtained by ridge-like regularised least squares with additive Gaussian noise of scale σ in the regressor,

$$\mathbf{W} = (\mathbf{X} + \mathbf{E})^+ \mathbf{Y}, \quad E_{ij} \sim \mathcal{N}(0, \sigma^2), \quad (2)$$

where $\mathbf{Y}$ is the time-aligned target and $(\cdot)^+$ the Moore–Penrose pseudoinverse. The noise scale σ acts as the ridge-regularisation parameter of the read-out fit (larger σ , stronger regularisation); $\sigma = 0.025$ unless swept (Fig. 3).

Closed-loop reconstruction. The model has two operating modes. In *teacher-forced* (open-loop) mode the real target drives the input, $\mathbf{u}(t) = f \hat{\mathbf{s}}(t)$; this mode is used to fit $\mathbf{W}$ and to compute the classifier features. In *closed-loop* (free-run) mode the read-out feeds back its own output, $\mathbf{u}(t) = f \mathbf{y}(t)$, generating an autonomous reconstruction. To validate the generative model (Fig. 2) the reservoir was reset and run closed-loop: the real target drove the network for the first 5 steps, after which it free-ran for the remainder of the session. The simulated reconstruction $\mathbf{Y}_{sim}$ and its (lagged) FC were compared to the empirical signals. Subject identifiability was quantified by the matrix of correlations between simulated subject i and empirical subject j , comparing the same-subject (diagonal) and cross-subject (off-diagonal) distributions. Both intervention families are evaluated consistently in both modes: a read-out interpolation ($\mathbf{W}_{int}$) and an input oscillation ($A \sin$) each alter the same reservoir, and each is scored by the classifiers on the teacher-forced FC-lag features and assessed for its departure from baseline on the closed-loop free-run FC (Fig. 5, figs. S5, S6); the two are therefore compared on a common footing, not one in open- and the other in closed-loop.

Read-out “G-space”. Each read-out was projected onto the row space of its own reservoir states: with $\mathbf{V}_k$ the top $K_{\text{PC}} = 200$ right singular vectors of $\mathbf{X}$, the projected read-out is $\widetilde{\mathbf{W}} = \mathbf{W}^\top \mathbf{V}_k \mathbf{V}_k^\top$, vectorised to $\mathbf{w}_i$. Stacking patients, centring, and taking the SVD of the centred matrix yielded patient coordinates (“G-scores”) in a low-dimensional read-out-archetype space; the first K components were used for classification. The projection rank $K_{\text{PC}} = 200$ and the low-dimensional sufficiency of the archetype basis are justified by a training/test reconstruction-error analysis over (K_{PC}, M) (fig. S2).

Lagged-FC features. From the reconstructed signals $\mathbf{Y} = \mathbf{W}^\top \mathbf{X}$ we computed lagged correlation matrices for lags 0–2 (zero-lag upper triangle plus full lag-1 and lag-2 matrices), concatenated into a feature vector per patient. A Gram-matrix SVD across patients (the transductive embedding) gave the “FC-lag” coordinates, of which the first K were used.

Classification. CC vs AD was classified with two algorithms on each embedding: a balanced Fisher linear discriminant (LDA; majority class sub-sampled to the minority within each training set) and a random forest (RF; 300 trees, $\sqrt{\cdot}$ features, minimum leaf size 3, balanced class weights). Performance was estimated by repeated stratified 5-fold cross-validation (80/20 train/test, 10 repeats), reporting balanced accuracy and the area under the ROC curve (AUROC) as the mean $\pm$ standard deviation over folds. We swept the embedding dimension $K \in [2, 50]$ and the read-out noise $\sigma \in [0.003, 0.5]$; reference slices were chosen by the marginal-mean AUROC across classifiers and the complementary axis. The SVD embeddings were built transductively (across all patients) while the classifiers were trained and tested on disjoint folds.

In-silico stimulation. The model was used as a counterfactual testbed by shifting each AD read-out toward the CC-mean read-out, $\mathbf{W}_{\text{int}} = (1 - \alpha) \mathbf{W}_{\text{AD}} + \alpha \bar{\mathbf{W}}_{\text{CC}}$, applied either to all 121 columns (sites; full-W) or to focal subsets. Here α is the read-out interpolation strength: $\alpha = 0$ leaves the unmodified AD read-out, $\alpha = 1$ replaces it entirely by the CC-mean read-out $\bar{\mathbf{W}}_{\text{CC}}$, and $\alpha > 1$ is over-correction. We nonetheless explore $\alpha > 1$ because the exact $\alpha = 1$ replacement can be insufficient: the reconstruction is closed-loop, so the reservoir state driven by the AD input keeps the FC partly AD-like even under the CC read-out, and crossing the decision boundary can require pushing past $\bar{\mathbf{W}}_{\text{CC}}$ (cf. fig. S5). (α is distinct from the oscillation amplitude A below.) Focal sites were ranked per patient by the column norm of $\Delta \mathbf{W} = \bar{\mathbf{W}}_{\text{CC}} - \mathbf{W}_{\text{AD}}$: the top-5, the top-1, and the top-1 site plus its 10 nearest neighbours in MNI space (parcel centroids from the atlas). For each α the perturbed read-out was re-scored by both classifiers, and the AD→CC reclassification

rate was the fraction crossing the CC/AD decision boundary. We report a heuristic reference amplitude of $\alpha = 4$ for summary comparisons; it is a convenient common operating point, not a biophysically calibrated limit, and the full dose-response is shown throughout.

Target selection (LDA-resonant). Stimulation targets were selected by their effect on the disease discriminant rather than by the magnitude of their read-out deviation. Each of the 121 sites was driven in turn by a sinusoid $A \sin(2\pi f_{\text{eig}} t)$ at the reservoir’s resonant frequency and a fixed reference amplitude ($A = 4$ in input units), injected during the teacher-forced pass with $\mathbf{W}$ unchanged, and the resulting reconstruction was rescored by the FC-lag classifier. The per-site *reduction* is the drop in FC-lag score relative to the unstimulated run, so that positive values move the patient toward the control distribution. Two targets follow: a single population-average target, the site maximising the reduction averaged over AD patients, and a *personalised* target, the site maximising the reduction for that individual. “LDA-resonant” refers to the personalised target unless stated otherwise.

Online closed-loop control. The closed-loop controller titrates the drive amplitude in real time at each patient’s fixed personalised LDA-resonant site, using only information available causally from the model’s own past output. The drive $A(t) \sin(2\pi f_{\text{eig}} t)$ begins after a short transient (5 steps) at $A = 6$, the amplitude used by the fixed open-loop protocol. Every 20 free-run steps the FC-lag score is recomputed from a sliding window of the preceding 20 reservoir states; if that score lies on the control side of the decision threshold the amplitude is decremented by 1 (floor 0), and if reclassification is lost it is incremented by 1 (ceiling 6). The controller therefore descends to the smallest dose that still maintains reclassification. No oracle access to the future trajectory or to the full-length reconstruction is used at any point. The reported dose is the mean amplitude over the run, and a patient counts as reclassified when the FC-lag score of the complete closed-loop reconstruction lies on the control side of the threshold.

Oscillatory stimulation. As an alternative focal protocol, a sinusoid $A \sin(2\pi f t)$ (amplitude A in input units, in the same units as $\mathbf{J}_{\text{in}} \mathbf{u}$ and distinct from the read-out interpolation strength α above) was injected as additional input at the top- k sites ($k = 1, 2, 5$) during the teacher-forced pass, with $\mathbf{W}$ unchanged; the oscillation propagates through the recurrent dynamics and reshapes the reconstructed FC, which was rescored by the FC-lag classifier. Two candidate drive frequencies were compared: the leading complex eigenmode of $\mathbf{J}$ (model-intrinsic, requiring the eigendecomposition of $\mathbf{J}$) and the peak of the empirical power

spectrum of the AD reservoir states (data-driven, obtained by FFT without inspecting $\mathbf{J}$); the two agree closely. Because $\mathbf{J}$ (and hence its eigenspectrum) is fixed and shared across subjects, the resonance f_{eig} is a property of the model, not a subject-specific fit. Frequencies were swept over $f \in [0, 0.5]$ cycles/step and amplitude up to *supra-physiological* values, i.e. beyond the RMS of the endogenous teacher-forcing drive (the target scaled by the feedback gain 0.1), so that the stimulated-to-baseline amplitude ratio $\text{RMS}_{\text{pert}}/\text{RMS}_{\text{base}} > 1$ (fig. S8). The state-level response of the linear reservoir to a periodic drive is governed by the transfer function $(i\omega\mathbf{I} - \mathbf{J})^{-1}\mathbf{J}_{\text{in}}\mathbf{b}$, the same linear response as in²⁵, which is large only when the drive frequency is near an eigenmode *and* the spatial input pattern $\mathbf{b}$ overlaps that eigenmode; a single site thus resonates cleanly at f_{eig} , whereas the summed pattern of several sites overlaps other modes and de-tunes the state-level resonance. Because the FC-lag score is a nonlinear functional of the read-out of the states rather than the excitation amplitude itself, this state resonance does not translate into a correspondingly sharp multi-site frequency optimum: we verified that selecting the five sites of maximal modal excitation $|\mathbf{w}^H\mathbf{J}_{\text{in}}|$ (with $\mathbf{w}$ the leading left eigenvector of $\mathbf{J}$) does *not* recover an f_{eig} optimum. The clean spectral prescription is therefore specific to single-site drive.

Leakage-free target selection. To test the therapy-target reclassification for classifier leakage we repeated the FC-lag analysis under leave-one-AD-patient-out cross-validation. For each held-out AD patient, the FC-lag embedding (a kernel PCA of the centred per-patient feature vectors), the balanced Fisher LDA, its decision threshold, and the site-selection criterion were all recomputed from the remaining patients (all CC plus the other AD patients); none of the held-out patient’s data entered the classifier. The held-out patient’s baseline and per-site resonantly-stimulated FC-lag features were then projected into this train-only space by out-of-sample kernel projection, its *personalised* site chosen as the argmax of the train-only score reduction over the 121 sites, and the *population-average* site as the argmax of the mean reduction across the training AD patients; reclassification was scored against the train-only threshold. Because the embedding projection is a Gram/kernel form, each fold is exact linear algebra on the base-patient Gram matrix, with the reservoir simulated once per (patient, site). We report the *net cure* rate: of the patients on the AD side of the train-only boundary at baseline, the fraction that stimulation moves to the CC side (fig. S7). The two targets bound the achievable effect from opposite sides: the *personalised* figure still reflects a per-patient argmax over the 121 candidate sites and should be read as an upper bound on achievable personalisation, whereas the *population-average* result, a single global site selected on the training patients alone, is the conservative,

fully out-of-sample estimate.

Physiological diagnostics. For every strategy and α we recorded the signal-amplitude ratio at stimulated sites, $\text{RMS}_{\text{pert}}/\text{RMS}_{\text{base}}$, and the mean absolute FC at those sites, to verify that reclassification (where it occurred) was achieved within plausible amplitude bounds.

Statistics and software. Group comparisons used two-sided Mann–Whitney U tests; distributions are shown as violins with medians or as mean $\pm$ SD/IQR as stated. Analyses were implemented in Python (NumPy, scikit-learn, nilearn, Matplotlib).

Ethics. This work is a secondary analysis of de-identified, publicly available human imaging data obtained from ADNI. The ADNI study was approved by the institutional review boards of all participating sites, and all participants (or their authorised representatives) provided written informed consent at enrolment; no new data were acquired for the present study.

Data and code availability. The imaging data analysed here were downloaded from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). ADNI data are available to qualified researchers upon registration and acceptance of the ADNI Data Use Agreement; in accordance with that agreement, the raw and derived imaging data are not redistributed here. The exact cohorts and sample sizes used in every analysis are listed in Table S1. All analysis and figure-generation code is openly available at github.com/cristianocapone/AD-reservoir-FC-stimulation.

Acknowledgments. Data used in preparation of this article were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf. Data collection and sharing for ADNI was funded by the ADNI (National Institutes of Health grant U01 AG024904) and DOD ADNI (Department of Defense award number W81XWH-12-2-0012). This work has been partially supported by the National Plan for Complementary Investments to the NRRP, project “D34H—Digital Driven

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580 **Author contributions.** C.C. designed the study, performed the analyses and in-silico experiments, and
581 wrote the manuscript. E.Ci. conceived the initial idea of a digital-twin model of Alzheimer’s disease
582 patients. E.Ce. downloaded and selected the original imaging data. A.C. and G.G. provided feedback on
583 the experiments and on the manuscript. L.F. contributed to defining the model. M.M., the senior author,
584 provided feedback on several of the experiments and on the manuscript. All authors read and approved the
585 final manuscript. (Author initials: C.C., Cristiano Capone; E.Ce., Enza Cece; L.F., Luca Falorsi; A.C.,
586 Andrea Ciardiello; G.G., Guido Gigante; E.Ci., Evaristo Cisbani; M.M., Maurizio Mattia.)

587 **Competing interests.** The authors declare no competing interests.

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Supplementary Information

Supplementary Note S1: Structural versus functional discrimination

On the same subjects and atlas, parcel-wise grey-matter volume classified AD from controls at AUROC ≈ 0.84 (linear discriminant analysis), exceeding both functional read-outs evaluated under an identical protocol on the same subjects (FC-lag and empirical tangent-space FC, AUROC ≈ 0.62 – 0.66). Combining functional with structural features did not improve on structure alone. This is consistent with atrophy being the stronger *diagnostic* signal; we nonetheless base the intervention analysis on the functional read-out, for the reasons given in the Introduction, because stimulation acts on dynamics rather than on tissue loss.

Table S1 | Cohorts used in each analysis. Minimally preprocessed resting-state fMRI (121 parcels, 140 volumes, TR = 3 s). “Subjects” are unique participants; “sessions” are individual scans (a subject may contribute several). Per-subject reservoir quantities use the first valid session.

Analysis	Controls (CC)	AD	Cross-validation / unit
Available pool	46 subj / 163 ses	34 subj / 104 ses	,
Data overview, FC similarity (Fig. 1)	163 sessions	104 sessions	per session
Reservoir fit & FC validation (Fig. 2); G-space & FC-lag classifiers (Fig. 3A–D); all in-silico stimulation (Figs 4–6, S2–S8)	38 subjects	33 subjects	one session/subject; 40 CC sessions sampled from the 48 verified CC subjects (38 passing the volume filter); repeated stratified CV
Cohort-size scaling (Fig. S1)	up to 46 subj	up to 34 subj	balanced subsets, leave-one-patient-out
Tangent-space SVM benchmark (Fig. 3E,F)	163 sessions (46 subj)	104 sessions (34 subj)	GSR preprocessing; nested 5×5 subject-grouped CV (267 sessions, train-only tangent reference)

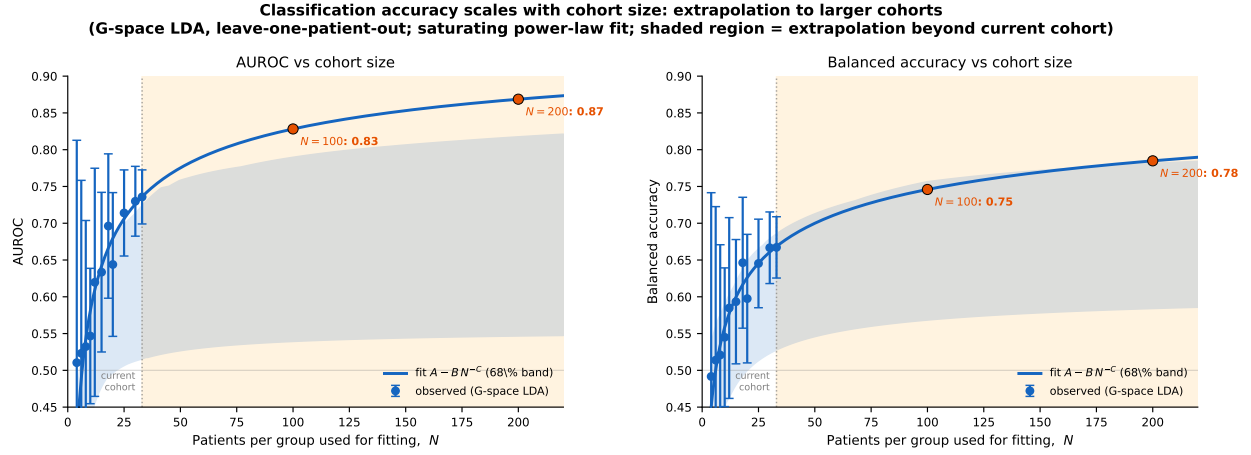


Fig. S1 | Classification accuracy scales with cohort size. G-space LDA balanced accuracy and AUROC (leave-one-patient-out) as a function of the number of patients per group used for fitting, N (blue points, mean $\pm$ SD over resampled subsets). A saturating power law $m(N) = A - BN^{-C}$ is fit to each metric (blue line; shaded 68% band from the fit covariance) and extrapolated beyond the current cohort (orange region). Accuracy has not saturated at $N = 33$: the extrapolation projects AUROC ≈ 0.83 at $N = 100$ and ≈ 0.87 at $N = 200$ (orange markers), indicating that the modest present accuracy is sample-size-limited rather than an intrinsic ceiling of the read-out representation.



Fig. S2 | Training and test reconstruction error of the read-out archetype model. Each fitted read-out W_i is projected onto the top- K_{PC} right singular vectors of its reservoir states, and the projected read-outs are then compressed across patients into M SVD archetypes; the error is the signal-space normalised mean-squared error, $NMSE = \langle (Y - XW_g^T)^2 \rangle / \langle Y^2 \rangle$, of the archetype reconstruction W_g ($\sigma = 0.025$; mean over 4 random 70/30 patient splits). (A) Training and (B) test NMSE over the (K_{PC} , M) grid. (C) Test NMSE vs M for each K_{PC} . Training error falls steeply with both K_{PC} and M , whereas test error is governed chiefly by K_{PC} (plateauing for $K_{PC} \gtrsim 100$) and is nearly flat in M beyond $M \approx 10$: additional archetypes fit training-specific variance, the widening train-test gap, without improving generalisation. This justifies the projection rank $K_{PC} = 200$ used throughout and shows that a low-dimensional archetype basis captures the cross-patient-shared read-out structure.

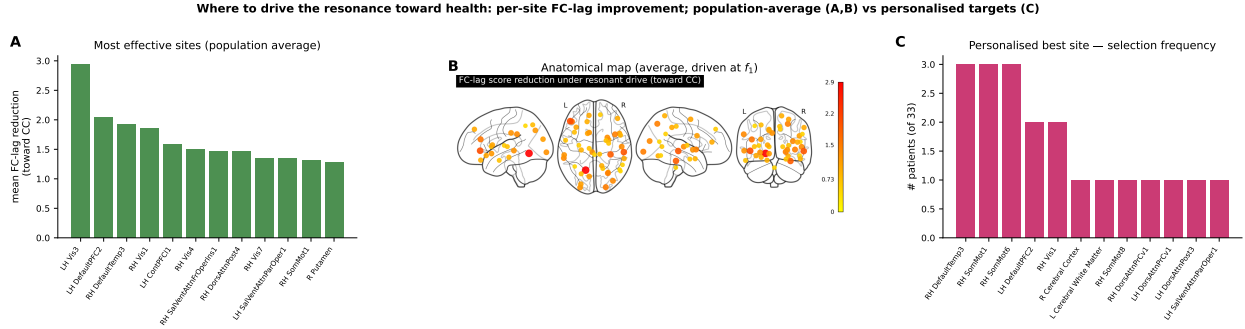


Fig. S3 | Where to drive the resonance toward health. Each of the 121 sites is driven at the leading eigenmode frequency f_1 (amplitude $A = 4$) and ranked by the resulting FC-lag LDA-score reduction toward CC, i.e. the projection of the resonant-stimulation effect onto the discriminant. **(A)** Population-average ranking of the most effective sites; the leading target is the left Default-network prefrontal cortex, followed by dorsal-attention, visual and somatomotor association cortex. **(B)** The same on a glass brain (marker size/colour = mean reduction): the effective sites are predominantly cortical and distributed. **(C)** When the target is chosen *per patient* (each subject's own model), the selected sites are heterogeneous, most are chosen by only a few patients and span cortical (Default, visual) and subcortical (hippocampus, brainstem, thalamus) structures, so no single global target is optimal for all, motivating personalised targeting.

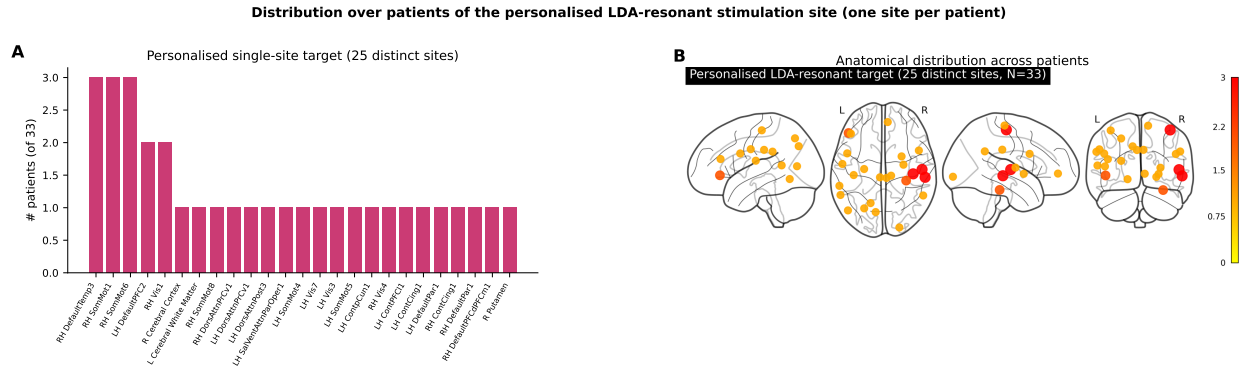


Fig. S4 | Per-patient top-1 stimulation site: comparison of the three selection criteria (same seeded reservoir; glass brain on top, selection-frequency histogram below; $N = 33$ AD patients). **(A)** ΔW (most relevant in the read-out; the single-site ΔW stimulation): the site of largest $\|W_{CC} - W_p\|$. These targets cluster on a few *subcortical* structures shared across patients (10 distinct sites; mostly accumbens, pallidum, brain-stem, amygdala), i.e. where the disease most distorts the read-out. **(B)** *Eigenmode*: the site of maximal coupling to the leading reservoir eigenmode. Being a property of the (shared) dynamics, it is a *single global site* for all patients (right dorsal-attention cortex). **(C)** *LDA-resonant* (per patient): the site whose resonant drive moves that subject's FC-lag discriminant most toward CC. These therapeutic targets are highly heterogeneous, **25 distinct sites for the 33 patients**, each selected by at most three, spanning Default-network, visual and attention cortex together with medial-temporal and subcortical structures. The three criteria answer different questions: where the disease lives in the read-out (ΔW), the fixed resonance node of the dynamics (eigenmode), and the personalised lever that drives recovery (LDA-resonant); only the last is genuinely individualised, so no single global site is adequate.

How much each intervention perturbs the simulation (vs the unstimulated model) for a given therapeutic effect: full exact ΔW vs personalised LDA-resonant vs eigenmode-site resonant

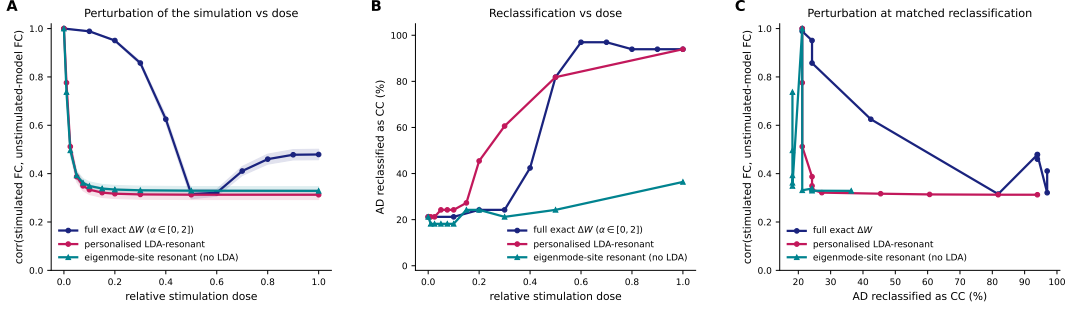


Fig. S5 | How much each intervention perturbs the simulation, for a given therapeutic effect: full exact ΔW , personalised LDA-resonant, and eigenmode-site resonant stimulation ($N = 33$ AD). The perturbation is the Pearson correlation (FC off-diagonals) between the *closed-loop free-run* simulated FC under stimulation (generative protocol of Fig. 2: drive 5 steps, then free-run) and the *unstimulated* model's own free-run FC, so it equals 1 at zero dose by construction and falls as the intervention reshapes the dynamics. (The unstimulated free-run reproduces the empirical FC at $r \approx 0.61$, as in Fig. 2.) Three interventions are compared: full-site ΔW interpolation $W_{\text{int}} = (1 - \alpha)W_p + \alpha W_{\text{CC}}$ ($\alpha \in [0, 2]$; $\alpha = 1$ full replacement by the CC-mean read-out, $\alpha > 1$ over-correction), the *personalised* LDA-resonant drive (each patient's own discriminant-aligned site), and the *eigenmode-site* resonant drive (single global site of maximal coupling to the leading eigenmode, selected by modal coupling *without* the LDA); both oscillatory drives use $A \in [0, 10]$ at f_1 , sampled finely at low A . **(A)** Perturbation versus normalised dose. Both single-site resonant drives fall *steeply but continuously* ($1 \rightarrow 0.74 \rightarrow 0.49 \rightarrow 0.38$ over $A = 0.1\text{--}0.75$) and saturate near $r \approx 0.33$: because the injected $A \sin$ is added to the un-scaled input whereas the data drive is $ff \cdot \text{tgt}$ ($ff = 0.1$), even $A = 1$ is an order of magnitude above the normal drive, and at f_1 the closed loop amplifies it until the resonant mode's (amplitude-normalised) spatial pattern dominates the FC. This perturbation is *network-wide*, not local to the driven node: recomputing the correlation after deleting the driven site's own row and column leaves it unchanged ($r = 0.32$ vs 0.32 at $A = 10$), i.e. every parcel, not just the stimulated one, becomes phase-locked to the driven mode. **(B)** Reclassification versus dose: the full ΔW correction reverts the FC-lag classifier only with over-correction ($\alpha \gtrsim 1.4$; $\leq 8\%$ at the exact $\alpha = 1$, since the reservoir state stays AD-driven, consistent with Fig. 4), reaching $\sim 90\%$ by $\alpha = 2$; the personalised LDA-resonant drive reaches 100% , whereas the eigenmode-site drive plateaus at $\sim 62\%$. **(C)** Perturbation at matched reclassification. The two single-site resonant drives perturb the simulation *near-identically* (correlation to the unstimulated model $\approx 0.32\text{--}0.34$ throughout), yet only the discriminant-aligned (LDA) target converts that same dynamical perturbation into full reclassification, selecting the site by its effect on the discriminant, rather than by modal coupling alone, extracts more therapeutic effect per unit of perturbation. The global ΔW overwrite perturbs less at low effect but must over-correct and rewrite all 121 read-out weights to compete.

Per-patient amplitude×frequency stimulation landscape: maximise reclassification, minimise distance to the unstimulated simulation

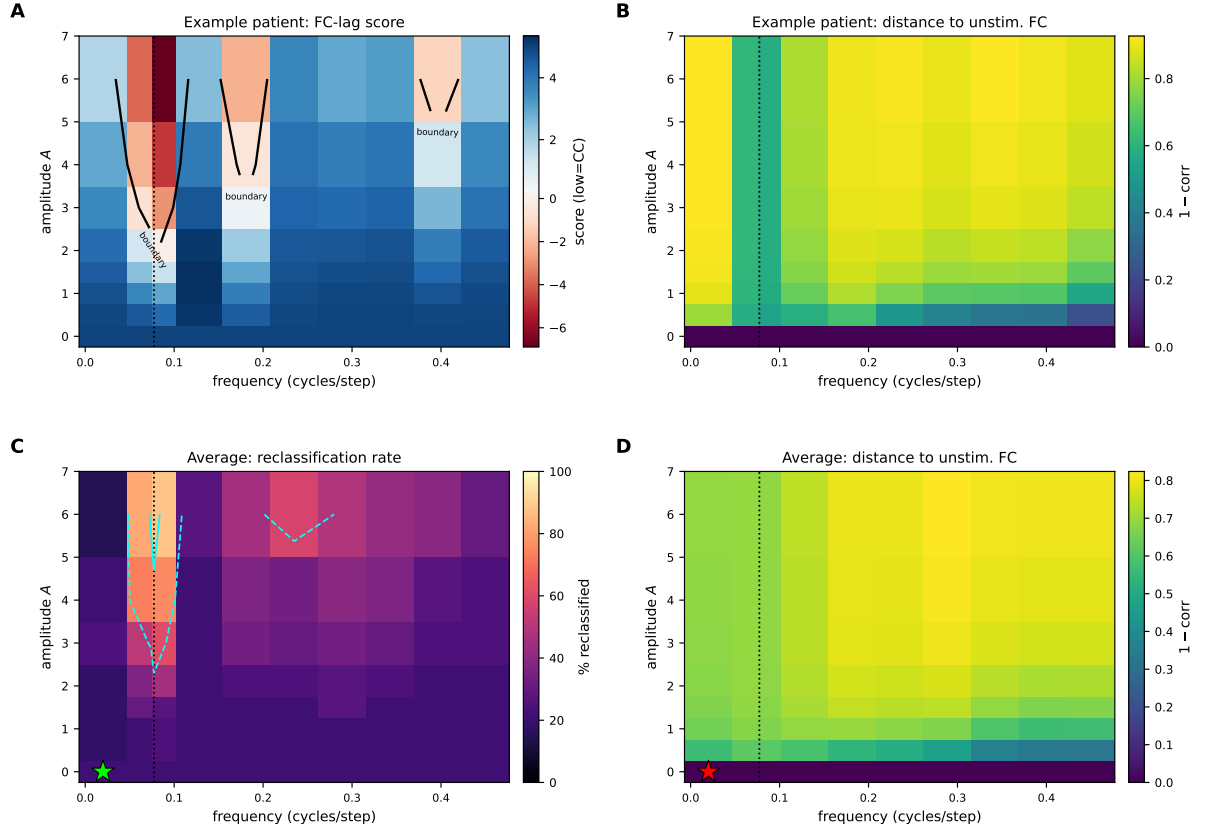


Fig. S6 | Per-patient amplitude×frequency stimulation landscape: trading off reclassification against departure from the unstimulated simulation ($N = 33$ AD). Each patient is driven at its personalised LDA-resonant site over an 8×10 grid of amplitude A and frequency f ; we record the FC-lag score (reclassified if below the boundary; classifier space) and the distance to the *unstimulated* simulation, $1 - \text{corr}$ between the stimulated and unstimulated closed-loop free-run FC. **(A,B)** One representative patient (median baseline score): the FC-lag score (A; boundary contour in black) and the distance (B). The score drops below the boundary only in a narrow band of frequencies around the reservoir resonance f_1 (dotted line). **(C)** Population reclassification rate: a narrow “tongue” centred on f_1 (cyan contours at 50% and 80%), at resonance patients reclassify at the *lowest* amplitude, whereas off-resonance requires much larger drive. **(D)** Population mean distance, which grows with amplitude and is itself elevated near f_1 . The two objectives therefore conflict: the operating point that maximises reclassification minus distance (★; $A = 6$, $f = f_1$, 100% reclassified) sits at high distance (≈ 0.68), and reverting most patients unavoidably requires a large perturbation (distance ≥ 0.66 , cf. fig. S5). Resonance is nonetheless the most *efficient* axis, the minimum-distance route to any target reclassification runs along f_1 , so the model’s prescription is to drive the personalised site at the intrinsic resonance, accepting the minimal perturbation that crosses the boundary.

The therapeutic reclassification effect survives a leakage-free (leave-one-AD-patient-out) discriminant (honest cross-validation raises the baseline misclassification; the stimulation effect is the increment over that baseline)

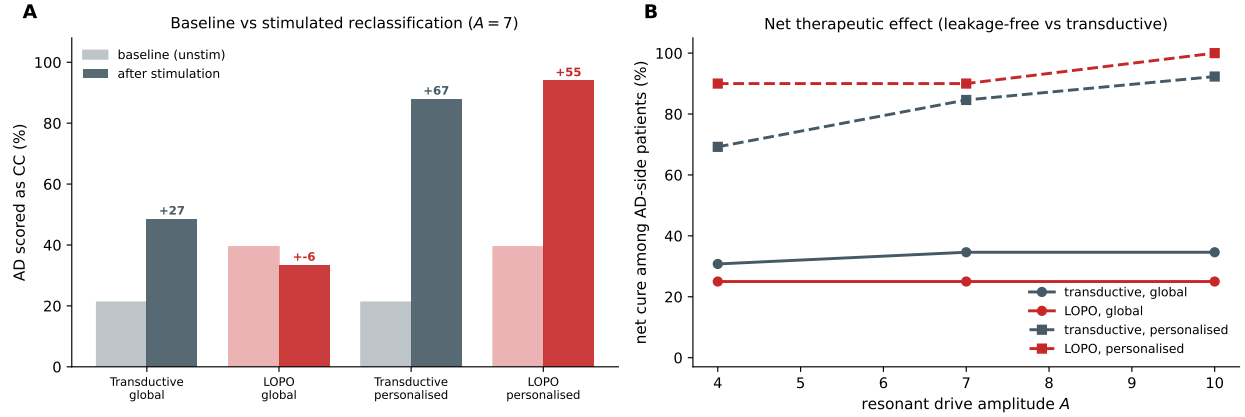


Fig. S7 | The therapeutic reclassification survives a leakage-free (leave-one-AD-patient-out) discriminant ($N = 33$ AD). The FC-lag embedding, the LDA discriminant, its decision threshold and the site-selection criterion are all rebuilt from the training patients only and applied to each held-out AD patient (Methods). Two targets are compared: the *population-average* site (one global site, selected on the training AD patients; conservative, no per-test-patient choice) and the *personalised* per-patient LDA-resonant site (an upper bound, a per-patient argmax over the 121 sites). (A) Fraction of AD patients scored as CC at baseline (unstimulated) versus after resonant stimulation at the reference amplitude $A = 7$, for each target under the transductive scheme (fit on all patients) and the leakage-free LOPO scheme (train-only); increments annotated. Honest cross-validation raises the baseline misclassification of AD as CC from 21% (transductive) to 39% (LOPO), so that only 20 of 33 AD patients lie on the AD side of the boundary unstimulated. (B) Net cure, the fraction of patients on the AD side at baseline that stimulation moves to the CC side, versus resonant drive amplitude, transductive vs LOPO, for the population-average (solid) and personalised (dashed) targets. Leakage-free, the single population-average target reaches $\sim 25\%$ net cure and the personalised target $\sim 90\%$ (100% at $A = 10$): the reclassification is a genuine, model-prescribed intervention effect, not an artefact of classifier leakage.

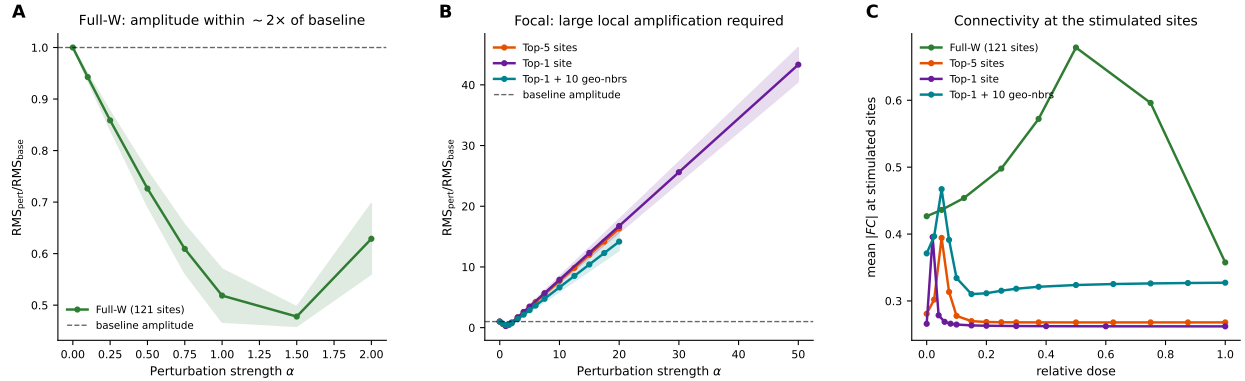


Fig. S8 | Physiological diagnostics of the read-out interpolation: distributed correction stays near the endogenous amplitude, focal correction does not ($N = 33$ AD). At the stimulated sites we record the signal-amplitude ratio $\text{RMS}_{\text{pert}}/\text{RMS}_{\text{base}}$ between the perturbed reconstruction $Y = \mathbf{XW}_{\text{int}}$ and the unperturbed one, and the mean $|FC|$ at those sites (Methods, “Physiological diagnostics”); mean $\pm 1\sigma$ over AD patients. (A) Full-W correction: the amplitude ratio remains within a factor of about two of baseline over the whole range $\alpha \in [0, 2]$ (dashed line, baseline), i.e. the distributed correction is achieved without driving the signal outside its endogenous range. (B) Focal corrections: the ratio instead grows approximately linearly with α , reaching $\sim 16\times$ (top-5) and $\sim 44\times$ (top-1) baseline at the amplitudes needed for even partial effects, i.e. deep into supra-physiological territory. (C) Mean $|FC|$ at the stimulated sites versus relative dose. Together these confirm that the failure of the focal strategies (Fig. 4C,F) is not a matter of insufficient dose: they are already operating far beyond plausible amplitude bounds.